

YIWEI YANG

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EDUCATION

UC Santa Cruz , Ph.D. Student	08/2022 – 06/2028
• Computer Science; advised by Andi Quinn; Fall 2022 Computer Architecture TA.	
ShanghaiTech University , Undergraduate	09/2018 – 06/2022
• Computer Science; coursework in compilers, networks, databases, OS, architecture, optimization, RL, and parallel computing; Spring 2021–22 Compiler TA.	

WORK EXPERIENCE

Zettai LLC , San Jose, CA	09/2025 – Present
(Founder) Founder	
• Founded Zettai LLC, focusing on compiler-driven heterogeneous XPU execution.	
Oxmiq Labs , Campbell, CA	06/2025 – 09/2025
(Compiler Engineer) Compiler Engineer Intern	
• Productionized HetGPU for compiler-driven heterogeneous GPU execution.	
Jump Trading , Shanghai, China	07/2020 – 09/2020
(Linux Team) Production Engineer Intern	
• Modeled high-frequency trading order books with linear regression.	
• Built a UI to automate job-to-CPU affinity assignment.	
• Developed an eBPF exporter for end-to-end GPFS OSS traces and bottleneck reporting.	

RESEARCH EXPERIENCE

Storage Systems Research Center, UC Santa Cruz	08/2022 – Present
(Graduate Research) Assistant	
• Designed data-driven OS policies for CXL.mem allocation, prefetching, and replacement. 🏆	
• Co-designed hardware and software for CXL.cache data movement. 📦	
• Built architecture-independent migration using WebAssembly and PTX. ^{MV} _{VM}	
• Built cross-kernel/userspace eBPF observability for distributed systems. 🐛	
• Extended vLLM/SGLang with CUDA/CXL attention kernels; optimized KV caching and continuous batching. 🧶	

SELECTED PUBLICATIONS

"SlugArch: A Boundary Replay Contract for Heterogeneous CXL Systems" Preprint Yiwei Yang , Andi Quinn.	
"WIO: Upload-Enabled Computational Storage on CXL SSDs" Preprint Yiwei Yang , Yanpeng Hu, Yusheng Zheng, Estabon Ramos, Jianchang Su, Andi Quinn, Wei Zhang.	
"CXLMemUring: Revolutionizing Memory Access Through Compiler-Driven Heterogeneous Execution" Preprint Yiwei Yang , Yusheng Zheng, Yuyi Li, Jianchang Su, Wei Zhang, Andi Quinn.	
"MVVM: Heterogeneous Checkpoint/Restore with Host-Specialized AOT Execution" Preprint Yiwei Yang , Aibo Hu, Yusheng Zheng, Brian Zhao, Xinqi Zhang, Andi Quinn.	
"TurnIP: Scaling Ternary Large Language Models" Preprint Ethan Sifferman, Eddy Abban, Yiwei Yang , Dustin Richmond.	
"Concordia: JIT-Compiled Persistent-Kernel Checkpointing for Fault-Tolerant LLM Inference" Preprint Yuhang Gan, Yiwei Yang , Yuyi Li, Xiangyu Gao, Yichen Wang, Rain Jiang, Xiaoning Ding, Andi Quinn, Chen Qian.	
"Safe and Programmable OS-Level GPU Resource Management with eBPF" ASPLOS27 Yusheng Zheng, Tong Yu, Yiwei Yang , Minghui Jiang, Xiangyu Gao, Jianchang Su, Yanpeng Hu, Wenan Mao, Wei Zhang, Dan Williams, Andi Quinn.	
"OCEAN: Open-source CXL Emulation at Hyperscale Architecture and Networking" SC26 Yiwei Yang , Mujahid Al Rafi, Kenza Bouasker, Zhen Peng, Xi Wang, Jesun Sahariar Firoz, Sayan Ghosh, Daniel Wong, Dong Li, Hyeran Jeon, Kevin Barker, Nathan Tallent, Andres Marquez and Luanzheng Guo.	
"hetGPU: Pushing pooling into heterogeneous GPU Domain" HCDS26 Yiwei Yang , Yusheng Zheng, Tong Yu, Andi Quinn.	
"CXLMemSim: Practical Performance Simulation and Characterization of CXL 3.0 Memory Systems" HPDC26 Yiwei Yang , Pooneh Safayenikoo, Jiacheng Ma, Tanvir Ahmed Khan, Andi Quinn.	
"Extending Applications Safely and Efficiently" OSDI25 Yusheng Zheng, Tong Yu, Yiwei Yang , Yanpeng Hu, Xiaozheng Lai, Dan Williams, Andi Quinn.	
"Kgent: Kernel Extensions Large Language Model Agent" eBPF24 Yusheng Zheng, Yiwei Yang , Maolin Chen, Andi Quinn.	

SKILLS

• Languages: Verilog, C++, Rust; working knowledge of Go, C, Java, and TypeScript.	
• Systems: Compilers and performance analysis; LLVM, MLIR, gem5, WASM, gdb, eBPF, PTX, QEMU, Chisel, Linux memory management and observability.	
• Machine Learning: ML and reinforcement learning; systems for LLMs and LLMs for systems.	

MISCELLANEOUS

• Interests: HPC, Architecture, Security, JIT, CXL, LLM4Sys4LLM, etc.	
• Led GeekPie_HPC to 2nd at SC-SCC21 and 4th at ISC22; advised Not-Slow-Slug to 2nd at ISC23.	
• Mentored Kiki Zhang and Zhen Cao toward Ph.D. study.	